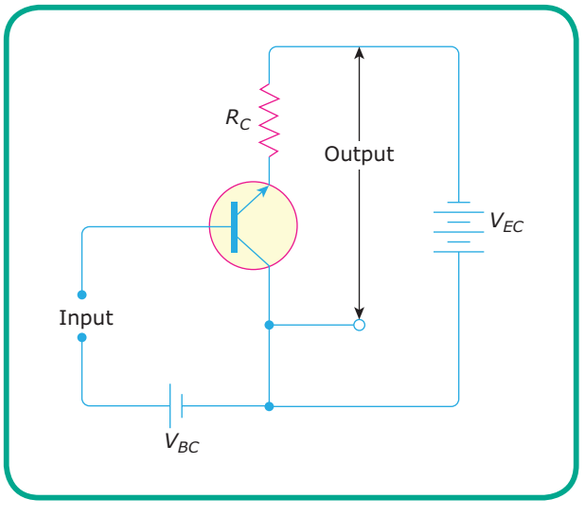


Figure 5.19 Common collector configuration

To determine the input characteristics, the emitter collector voltage V_{EC} is kept constant at zero and base current I_B is increased from zero by increasing V_{BC}. This is repeated for higher fixed values of V_{EC}. A curve is drawn between base current and base emitter voltage at constant collector base voltage is shown in figure 5.20.

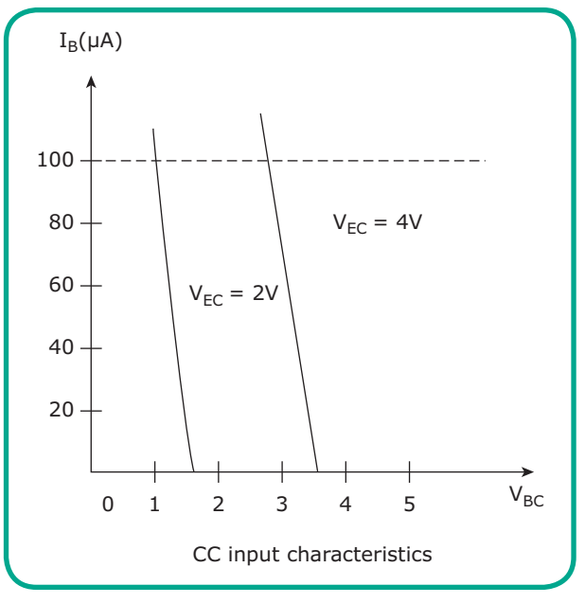


Figure 5.20 Input characteristics

Output Characteristics

It is defined as the characteristic curve drawn between output voltage to output current whereas input current is constant.

To determine output characteristics, the base current I_B is kept constant at zero and emitter current I_E is increased from zero by increasing V_{EC}. This is repeated for higher fixed values of I_B.

From the characteristic it is seen that for a constant value of I_B, I_E is independent of V_{EC} and the curves are parallel to the axis of V_{EC} as shown in the figure 5.21

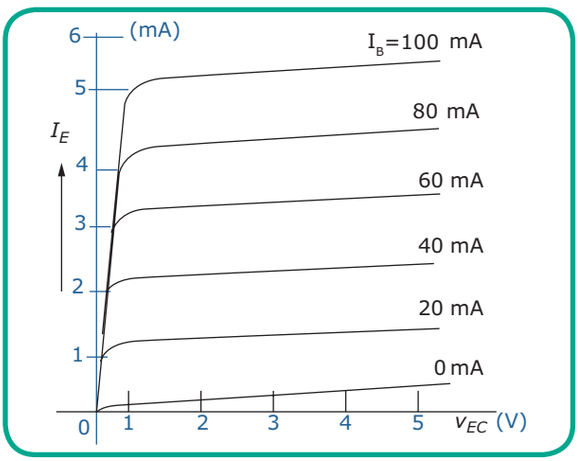


Figure 5.21 Output characteristics

Current Amplification Factor(γ).

It is the ratio of output current to input current. In common collector connection, the input current is the base current I_B and output current is the emitter current I_E.

The ratio of change in emitter current to the change in base current at constant emitter collector voltage V_{EC} is known as current amplification factor i.e.

$$\gamma = \frac{\Delta I_E}{\Delta I_B} \text{ at constant } V_{EC}$$

5.10 MULTISTAGE TRANSISTOR AMPLIFIER

A transistor circuit containing more than one stage of amplification is known as multistage transistor amplifier.

In a multistage amplifier, a number of single amplifiers are connected in cascade

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